

+3V3
 C19
 10uF/6.3V/20%/X5R/C0402
 GND

+1V1
 VREG_LX
 L1
 7.3uH/20%/Isat2.4A/DCR140m
 C21
 4.7uF/10V/20%/X5R/C0402
 GND

VREG_AVDD
 R5
 150/100/1% 0402
 GND

XIN
 Y1
 12MHz/CL10pF/ESP8266/ABM8-272-T3
 C22
 15pF/50V/5%/C0G/C0603
 GND

XOUT
 R6
 1k
 GND

BOOTSEL
 SW1
 SW
 GND

RESET
 SW2
 RST
 GND

SWDIO
 SWCLK
 J2
 SWD TC2030
 GND

ESP8266

. NOT near L1/C21.

L1 polarity dot (pad 1) must face VREG_LX.
 Do not substitute L1 or Y1 – see README.

The schematic shows three components: a 3V3 voltage regulator, a 3V3 VTGT voltage source, and a 3V3 VTT voltage source. The 3V3 voltage regulator is connected to GND and has a 100nF capacitor. The 3V3 VTGT voltage source is connected to GND and has a 100nF capacitor. The 3V3 VTT voltage source is connected to GND and has a 100nF capacitor.

The main circuit consists of two SN74LVC1T45DBV receivers (U4 and U6) and one SN74LVC1T45DBV driver (U5). The SN74LVC1T45DBV receiver (U4) is connected to the 3V3 VTGT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. The SN74LVC1T45DBV receiver (U6) is connected to the 3V3 VTT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. The SN74LVC1T45DBV driver (U5) is connected to the 3V3 VTGT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.

The SN74LVC1T45DBV receiver (U4) is connected to the 3V3 VTGT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. The SN74LVC1T45DBV receiver (U6) is connected to the 3V3 VTT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. The SN74LVC1T45DBV driver (U5) is connected to the 3V3 VTGT voltage source and has pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.

Diagram illustrating the connection of the PWR_FLAG pin to different power sources for H1, H2, H3, and H4 mounting holes:

- H1 MountingHole:** PWR_FLAG is connected to GND.
- H2 MountingHole:** PWR_FLAG is connected to VBUS.
- H3 MountingHole:** PWR_FLAG is connected to +5V.
- H4 MountingHole:** PWR_FLAG is connected to +1V1.

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